AP-E6 Deterministic Classical Gate: A Coupled Relaxed $B = 1$ Solution, Multigrid Scan, and Exact Sparse $n + s$ Projected-Curvature Audit

Route F research note

16 July 2026

Abstract

We replace the analytic-profile and single-field limitations of the preceding charged two-colour proxy by a coupled relaxation. The declared classical sector contains a unit meson field $n : \mathbb{R}^3 \to S^3$, a neutral breathing field s, a charge-two complex diquark at zero background, and free gauge–ghost quadratic blocks. A two-field hedgehog $(F(r), s(r))$ is solved as a nonlinear boundary-value problem on four radial volumes. The same relaxed solution is sampled on three lattice spacings and three physical volumes. Starting from one explicitly stated finite-difference energy, we assemble its analytic sparse embedding Hessian, apply the constrained Riemannian correction, restrict it to T_nS^3, and remove three translations and three isorotations by an explicit orthogonal projector. Toron and constant-ghost projectors are audited separately. Negative controls detect a charged tachyon, an omitted collective direction, and loss of Derrick stationarity without the Skyrme term.

The cubic-grid audit deliberately exposes an obstruction. The sampled continuum solution has tangential gradient density approximately 0.35–0.43, while the displayed lattice baryon estimator only rises from 0.168 to 0.625. It is therefore off shell for the finite cubic energy. Its negative projected curvatures cannot be called physical instability eigenvalues; instead they fail-close the discrete-stationarity and topology-convergence gates.

Completeness below refers only to exact differentiation of the declared deterministic classical sector. No fermion determinant, interacting colour-link ensemble, four-dimensional importance sampling, HMC, or quantum continuum limit is computed. Consequently the microscopic charged-QC2D lane and the degree-one Route-E portal remain closed.

Contents

1 Scope and epistemic levels

Two-colour QCD has pseudoreal matter and genuine meson–diquark competition [1]. Recent linear and extended-linear sigma analyses make clear that scalar and spin-one channels can reorganise in dense matter [2, 3]. A classical EFT soliton is therefore neither a lattice measurement nor a proof of the microscopic phase. Four-dimensional simulations exist on much larger ensembles [4]; a recent tensor calculation controls a different one-dimensional regulator [5]. Neither supplies the charged scalar, soliton background, or projected Hessian studied here.

We distinguish four statements:

(L1) a stationary solution of an explicitly declared classical EFT;

(L2) the exact second derivative of a declared discretisation, including its collective-coordinate quotient, with a separate stationarity test;

(L3) a dynamical four-dimensional Euclidean path integral;

(L4) a regulator-independent quantum continuum phase.

This note reaches L1 only in the coupled hedgehog sector; no independent principle-of-symmetric-criticality proof is invoked to promote it to arbitrary three-dimensional variations. It reaches the algebraic differentiation part of L2. The sampled cubic configuration fails the discrete-stationarity and topology tests, so its L2 operator is an off-shell curvature diagnostic.

Fail-closed gate 1.1 (Meaning of "complete")**.** Sparse differentiation is complete only for the $n+s$ block: every entry is a derivative of one energy, and a geodesic energy difference checks it. The Δ spectrum is a separate Dirichlet block, while the gauge/ghost algebra uses a separate periodic 5^3 grid. They are not one same-grid assembled Hessian. Moreover, exact differentiation does not make the sampled cubic field stationary. Thus the aggregate `sparse_projected_hessian_complete_in_declared_sector` gate is false, and this is not yet a physical stability Hessian nor the complete gauge–meson–ghost–fermion Hessian of charged QC2D. In particular, there is no Wilson/overlap fermion determinant on the same relaxed background.

2 Declared coupled classical EFT

2.1 Fields and energy

Let $n = (n_0, \boldsymbol{n}) \in S^3$, $n \cdot n = 1$, and let $s \in \mathbb{R}$ be a neutral breathing field with vacuum value one. In units $f = e = 1$, define

$$E[n,s] = \int_{\mathbb{R}^3} \mathrm{d}^3x \left\{ \frac{1}{2}\partial_i n \cdot \partial_i n + \frac{\kappa}{2} \sum_{i<j} \left[|\partial_i n|^2 |\partial_j n|^2 - (\partial_i n \cdot \partial_j n)^2 \right] + \frac{1}{2}|\nabla s|^2 + \frac{1}{2}m_s^2 (s-1)^2 + \beta^2 s^2 (1-n_0) \right\}. \tag{1}$$

The first line is the sigma-plus-Skyrme energy [6, 7]. The last term couples the breathing field to explicit chiral breaking while preserving the vacuum $(n, s) = (\mathbf{1}, 1)$. It is a declared diagnostic interaction, not a UV matching result. The benchmark is

$$\beta = \frac{1}{2}, \qquad m_s = \frac{5}{2}, \qquad \kappa = 1. \tag{2}$$

A charge-two complex scalar is retained only to quadratic order about $\overline{\Delta} = 0$:

$$\mathcal{H}_\Delta = -D_i^{(2)} D_i^{(2)} + m_\Delta^2 + \chi(1 - \overline{n}_0), \qquad m_\Delta^2 = 0.81, \quad \chi = -0.30. \tag{3}$$

At zero charged background, gauge–diquark mixing vanishes. This does not cover a Higgsed/condensed branch.

2.2 Topology and the coupled hedgehog

Use

$$n_0 = \cos F(r), \qquad \boldsymbol{n} = \widehat{\boldsymbol{x}} \sin F(r), \qquad s = s(r), \tag{4}$$

with $F(0) = \pi$, $F(\infty) = 0$, $s'(0) = 0$, and $s(\infty) = 1$. Then

$$B = -\frac{1}{2\pi^2} \int \det(n, \partial_1 n, \partial_2 n, \partial_3 n)\, \mathrm{d}^3 x = -\frac{2}{\pi} \int_0^\infty \sin^2 F\, F'\, \mathrm{d}r = +1. \tag{5}$$

Substitution into Eq. (1) gives $E = 4\pi \int \mathcal{E}_r \mathrm{d}r$, where the dimensionless radial integrand is

$$\begin{aligned}\mathcal{E}_r = {} & \frac{1}{2} r^2 F'^2 + \sin^2 F + \kappa \left(\sin^2 F F'^2 + \frac{\sin^4 F}{2r^2} \right) + \frac{1}{2} r^2 s'^2 \\ & + \frac{1}{2} m_s^2 r^2 (s-1)^2 + \beta^2 r^2 s^2 (1 - \cos F).\end{aligned} \tag{6}$$

2.3 Euler–Lagrange boundary-value problem

Direct variation gives

$$(r^2 + 2\kappa \sin^2 F) F'' = -2rF' - 2 \sin F \cos F (\kappa F'^2 - 1) + \frac{2\kappa \sin^3 F \cos F}{r^2} + \beta^2 r^2 s^2 \sin F, \tag{7}$$

$$s'' + \frac{2}{r} s' = m_s^2 (s-1) + 2\beta^2 s (1 - \cos F). \tag{8}$$

Near the origin,

$$F = \pi - ar + br^3 + O(r^5), \qquad b = \frac{a(2\kappa a^4 + 4a^2 + 3\beta^2 s_0^2)}{30(1 + 2\kappa a^2)}, \tag{9}$$

and

$$s = s_0 + \frac{1}{6}[m_s^2 (s_0 - 1) + 4\beta^2 s_0] r^2 + O(r^4). \tag{10}$$

These series, rather than a hard-coded profile, seed the nonlinear BVP.

Proposition 2.1 (Generalised Derrick identity)**.** *Every sufficiently localised stationary solution of Eq.* (6) *satisfies*

$$E_2 + E_{s,\mathrm{kin}} - E_4 + 3(E_{s,\mathrm{pot}} + E_m) = 0. \tag{11}$$

Its scale curvature is $2E_4 + 6(E_{s,\mathrm{pot}} + E_m)$.

Proof. Apply $F(r) \mapsto F(r/\lambda)$, $s(r) \mapsto s(r/\lambda)$. Two-derivative terms scale as λ, the Skyrme term as λ^{-1}, and potential terms as λ^3. Differentiation at $\lambda = 1$ proves Eq. (11); a second derivative gives the stated curvature. □

3 One lattice energy and its exact sparse Hessian

3.1 Declared discretisation

On an open cubic lattice of spacing a, freeze the boundary to the sampled continuum profile rather than overwriting it by the vacuum, let $\delta_i n_x = (n_{x+\hat{\imath}} - n_x)/a$. The single energy used by the implementation is

$$E_a = a^3 \sum_{x,i} \frac{1}{2}(|\delta_i n_x|^2 + |\delta_i s_x|^2) + \frac{\kappa a^3}{2} \sum_{x,i<j} \left(|\delta_i n_x|^2 |\delta_j n_x|^2 - (\delta_i n_x \cdot \delta_j n_x)^2\right)$$
$$+ a^3 \sum_x \left[\frac{1}{2} m_s^2 (s_x - 1)^2 + \beta^2 s_x^2 (1 - n_{0,x})\right]. \quad (12)$$

No quadratic block is inserted by hand: every meson–breathing entry below is an analytic second derivative of Eq. (12). This sampled-profile Dirichlet condition is part of the off-shell diagnostic. For the fixed-spacing volume scan its maximum boundary n-distance from the vacuum decreases from about 0.350 at $L = 4$, through 0.120 at $L = 6$, to 0.0493 at $L = 8$; it is not silently treated as an exact vacuum wall.

For a Skyrme cell pair $u = \delta_i n$, $v = \delta_j n$, write $W = \kappa(|u|^2|v|^2 - (u \cdot v)^2)/2$. The local blocks are

$$W_{uu} = \kappa(|v|^2 \mathbf{1} - vv^T), \qquad W_{vv} = \kappa(|u|^2 \mathbf{1} - uu^T),$$
$$W_{uv} = \kappa[2uv^T - vu^T - (u \cdot v)\mathbf{1}], \qquad W_{vu} = W_{uv}^T. \quad (13)$$

Together with nearest-neighbour Laplacians and the local s–n_0 cross derivative, Eq. (13) produces a sparse symmetric embedding Hessian H_{emb}.

3.2 Constrained tangent Hessian

At each site choose $E_x : \mathbb{R}^3 \to T_{n_x} S^3$,

$$E_x^T E_x = \mathbf{1}_3, \qquad E_x^T n_x = 0. \quad (14)$$

If $g_x = \partial E_a / \partial n_x$ and $\lambda_x = n_x \cdot g_x$, the constrained Riemannian Hessian is

$$H_{\mathrm{tan}} = T^T \left[H_{\mathrm{emb}} - \bigoplus_x \lambda_x \mathbf{1}_4\right] T, \qquad T_x = \begin{pmatrix} E_x & 0 \\ 0 & 1 \end{pmatrix}. \quad (15)$$

The multiplier term is essential: simply restricting the ambient Hessian is not the second variation on the sphere. Equation (15) is assembled analytically as CSR sparse data.

An independent test chooses a horizontal vector v, follows the sitewise geodesic

$$n_x(t) = \cos(t|v_x|) n_x + \frac{\sin(t|v_x|)}{|v_x|} v_x,$$

and evaluates

$$\frac{E_a[n(t), s + tv_s] - 2E_a[n, s] + E_a[n(-t), s - tv_s]}{t^2} = v^T H_{\mathrm{tan}} v + O(t^2). \quad (16)$$

This checks the Weingarten/Lagrange-multiplier term as well as the ambient second derivative. It checks differentiation, not stationarity.

4 Collective and gauge projectors

4.1 Translations and isorotations

Let the six sampled collective vectors be

$$z_i = (E^T \partial_i n, \partial_i s), \quad i = 1, 2, 3, \qquad z_{3+a} = (E^T(0, \boldsymbol{e}_a \times \boldsymbol{n}), 0), \quad a = 1, 2, 3. \tag{17}$$

After a thin QR decomposition $Z = QR$, $Q^T Q = \mathbf{1}_6$, define

$$P_{\text{coll}} = \mathbf{1} - QQ^T, \qquad H_{\text{proj}} = P_{\text{coll}} H_{\tan} P_{\text{coll}}\big|_{\operatorname{im} P_{\text{coll}}}. \tag{18}$$

The implementation tests $P^2 = P$, $Q^T P = 0$, eigenvector horizontality, and an intentionally incomplete rank-five projector. Because a finite cube and its Dirichlet boundary lift continuum translations, raw collective Rayleigh quotients are reported rather than declared exact zeroes.

4.2 Gauge and ghost zero modes

For the decoupled free Abelian block on a periodic three-dimensional audit grid, let G be the forward gradient and $L = G^T G$. Covariant gauge fixing gives

$$K_A(\xi) = \mathbf{1}_3 \otimes L + (\xi^{-1} - 1) G G^T, \qquad M_{\text{FP}} = L. \tag{19}$$

The three constant link torons and one constant ghost are removed by

$$P_A = \mathbf{1} - \sum_{i=1}^{3} t_i t_i^T, \qquad P_c = \mathbf{1} - \frac{\mathbf{1}\mathbf{1}^T}{V}. \tag{20}$$

Then, up to a background-independent normalization,

$$\frac{1}{2} \log \det{}' K_A(\xi) - \log \det{}' M_{\text{FP}} + \frac{V-1}{2} \log \xi \tag{21}$$

is ξ-independent [8].

5 Numerical certificate

5.1 Coupled radial relaxation

The final tables and exact values are machine-generated in `route_f/output/ap_e6_relaxed_b1_multigrid_hessian.json`. The BVP is solved independently at $R = 8, 10, 12, 16$. The volume sequence converges to approximately

$$E/(4\pi f/e) = 6.1282155, \qquad s(0) = 0.9171290, \qquad B = 1, \tag{22}$$

with BVP residual below 3×10^{-7}. The nonconstant $s(r)$ and the change between boxes demonstrate a genuine coupled relaxation rather than evaluation of the old analytic profile.

5.2 Multigrid, multivolume, and spectra

The fixed-volume scan uses $L = 8$ at $N = 9, 13, 17$; the fixed-spacing scan uses $(N, L) = (9, 4), (13, 6), (17, 8)$, all with $a = 0.5$. Every row re-samples the same $R = 16$ relaxed continuum solution and reassembles the exact Hessian of Eq. (12). We report the lattice topological estimator, sparse dimension and nonzero count, tangential stationarity residual, raw low spectrum, projected spectrum, and charge-two diquark spectrum.

The $N = 17$ reduced block has 13500 variables and 395538 nonzero entries. Its symmetry residual is below 10^{-16}; the best relative residual in Eq. (16) is below 2×10^{-9}. Projector idempotency,

N	a	B_a	$\lVert \operatorname{grad} E_a \rVert_{2,\mathrm{dens}}$	$\lambda_{\mathrm{proj},0}$	$\lambda_{\Delta,0}$
9	1.000000	0.168018	0.34766	-10.79222	1.219089
13	0.666667	0.440012	0.43018	-23.42272	1.223574
17	0.500000	0.625354	0.42171	-28.44285	1.224884

Table 1: Fixed-$L = 8$ off-shell curvature scan. The improving but still coarse B_a and non-small gradient prohibit a stability interpretation.

horizontality, and eigenvector-overlap residuals are respectively below 2×10^{-16}, 10^{-13}, and 10^{-8}. Thus the negative values in Table 1 are not caused by omitting the sphere-curvature term or by collective leakage. Yet they remain off-shell finite-grid curvatures, because the background has not been re-relaxed for Eq. (12).

At fixed $a = 0.5$, the negative branch is volume-stable: $-28.41847, -28.44005, -28.44285$. The lowest diquark box state is unbound and moves as $1/L^2$; a linear fit extrapolates to 0.813709, close to the declared threshold $m_\Delta^2 = 0.81$, with RMS residual 1.28×10^{-3}. These are algebraic finite-grid controls. Since neither cubic stationarity nor $B_a \to 1$ is demonstrated, the top-level `multigrid_multivolume_converged` flag is false.

For downstream same-soliton work, the public function `solve_relaxed_profile(box=16.0, sample_points=4097)` returns canonical arrays (r, F, s). The JSON records their little-endian float64 SHA-256, so a Callias/Yukawa calculation can prove that it consumed this exact solution.

5.3 Negative controls

Three falsifiers are mandatory.

(N1) Replacing $\chi = -0.30$ by $\chi = -5$ must create a negative charge-two scalar eigenvalue.

(N2) Removing one column from Q must leave unit leakage along the omitted collective direction.

(N3) Setting $\kappa = 0$ leaves the scale derivative $E_2 + E_{s,\mathrm{kin}} + 3(E_{s,\mathrm{pot}} + E_m) > 0$, so no finite-scale Derrick saddle exists.

6 What is and is not closed

The following finite classical statements are closed by the verifier:

1. a coupled, regular, finite-volume-converged $B = +1$ BVP exists in the declared EFT;
2. the exact analytic sparse $n + s$ Hessian of the declared lattice energy is symmetric and its tangent constraint term is included;
3. translation, isorotation, toron, and constant-ghost projectors satisfy their respective algebraic tests, with the different boundary/grid choices recorded rather than merged;
4. the geodesic directional second difference validates the analytic Hessian, the diquark block remains positive, and all three negative controls fire.

The following remain false:

1. stationarity of the sampled field for the cubic energy, $B_a \to 1$, and a physical projected stability spectrum; the observed negative curvatures are consequently off-shell diagnostics;
2. one same-grid $n + s + \Delta$+gauge/ghost assembled sparse Hessian;
3. the full gauge–meson–ghost–fermion Hessian on one microscopic action;
4. interacting $SU(2)_c$ links, the fermion determinant, and measure positivity;

5. four-dimensional importance sampling or HMC;

6. a continuum quantum phase, finite-amplitude global stability, and a microscopic derivation of m_s, m_Δ, χ;

7. permission to construct the degree-one Route-E portal.

Fail-closed gate 6.1 (Lane status)**.** The calculation advances the classical soliton/Hessian lane but does not close it at the microscopic level. The exact machine flag names are retained in the JSON certificate. In prose, the false gates are:

- discrete stationarity, lattice-topology convergence, and a physical projected Hessian at a stationary discrete solution;
- multigrid/multivolume convergence, aggregate same-grid assembly, and the full gauge–meson–ghost–fermion Hessian;
- dynamical four-dimensional sampling and a continuum quantum phase;
- physics promotion and closure of this lane.

7 Next decisive computations

1. Put Wilson or overlap fermions, dynamical $SU(2)_c$ and Abelian links, and the charged scalar on one four-dimensional Euclidean action; then sample it rather than freezing the background.

2. Re-relax the soliton on every four-dimensional ensemble, compute the fermion determinant curvature or stochastic estimator, and extrapolate in a, spatial volume, Euclidean time, and statistics.

3. Test finite-amplitude escape paths in the faithful target; a positive local projected Hessian is only a necessary condition.

References

[1] John B. Kogut, Mikhail A. Stephanov, Dominique Toublan, Jacobus J. M. Verbaarschot, and Ariel Zhitnitsky. QCD-like theories at finite baryon density. *Nuclear Physics B*, 582:477–513, 2000.

[2] Daiki Suenaga, Kotaro Murakami, Etsuko Itou, and Kei Iida. Probing the hadron mass spectrum in dense two-color QCD with the linear sigma model. *Physical Review D*, 107:054001, 2023.

[3] Daiki Suenaga, Kotaro Murakami, Etsuko Itou, and Kei Iida. Mass spectrum of spin-one hadrons in dense two-color QCD: Novel predictions by extended linear sigma model. *Physical Review D*, 109:074031, 2024.

[4] A. Begun, V. G. Bornyakov, V. A. Goy, A. Nakamura, and R. N. Rogalyov. Study of two color QCD on large lattices. *Physical Review D*, 105:114505, 2022.

[5] Kwok Ho Pai, Shinichiro Akiyama, and Synge Todo. Two-color lattice QCD in (1+1) dimensions with Grassmann tensor renormalization group. *arXiv preprint*, 2025.

[6] T. H. R. Skyrme. A nonlinear field theory. *Proceedings of the Royal Society A*, 260:127–138, 1961.

[7] Gregory S. Adkins, Chiara R. Nappi, and Edward Witten. Static properties of nucleons in the Skyrme model. *Nuclear Physics B*, 228:552–566, 1983.

[8] L. D. Faddeev and V. N. Popov. Feynman diagrams for the Yang–Mills field. *Physics Letters B*, 25:29–30, 1967.